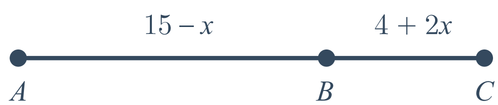
Proving segment relationships

1 Determine whether each of the following statement is sometimes, always, or never true.

- **a** The segments $\overline{AB}$ and $\overline{BA}$ describe the same set of points.
- **b** Given A and B are two different points, the rays $\overrightarrow{AB}$ and $\overrightarrow{BA}$ describe the same set of points.
- **c** Given A and B are two different points, line $\overleftrightarrow{AB}$ and line $\overleftrightarrow{BA}$ describe the same set of points.
- **d** Given A, B, and C are three different points, the rays $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are opposite rays.

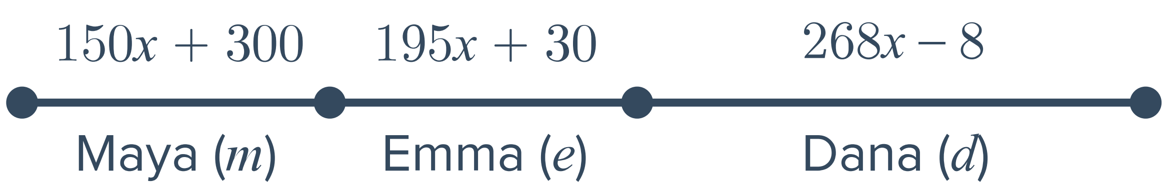
2 Given line $\overleftrightarrow{AB}$ passes through point C, is it true that C must also lie on $\overrightarrow{AB}$?

3 Suppose that points A, B, and C are collinear, with point B between points A and C.



Solve for x if $AC = 21$, $AB = 15 - x$, and $BC = 4 + 2x$.

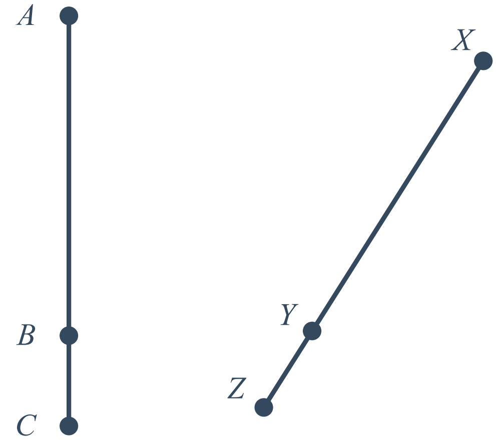
4 Maya, Emma and Dana are running in a relay race. Each will run a segment of the track until the last member gets to the finish line. The total length of the race (t) is 4000 m. Maya will run $150x + 300 \text{ m}$, Emma will run $195x + 30 \text{ m}$, and Dana will run $268x - 8 \text{ m}$.



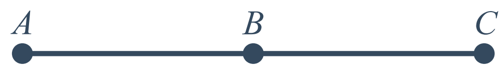
Using the information provided, solve for x.

5 Given: P, Q, R, and S are collinear
Prove: $PQ = PS - QS$

6 In the diagram shown, A, B, and C are collinear, and X, Y, and Z are collinear.
Given: $\overline{AB} \cong \overline{XY}$ and $\overline{BC} \cong \overline{YZ}$
Prove: $\overline{AC} \cong \overline{XZ}$



7 Given: B is the midpoint of $\overline{AC}$
Prove: $2AB = AC$



8 In the diagram shown, points R, S, T, and U are collinear.
Given: $\overline{RT} \cong \overline{SU}$
Prove: $\overline{RS} \cong \overline{TU}$